SUNY Old Westbury 2026-2027 Undergraduate Catalog

Physics, B.S.

The Physics B.S. program prepares students for a wide variety of careers and professions by providing a universal skill set in quantitative problem solving, analytical reasoning, and mathematical modeling. Students hone these skills by investigating the fundamental laws of nature and exploring deep questions about the properties of matter, the origin and fate of the universe, and nature of space and time.

The Physics B.S. program is designed to meet the diverse needs of physics majors by offering four concentrations: a general physics concentration designed to prepare students for graduate studies and three career-oriented concentrations. All bachelor's students will take the physics core of foundational and upper division courses. Further advanced courses are determined by the choice of concentration.

General Physics concentration: A traditional physics B.S. program designed for students continuing to graduate school or pursuing a career in STEM.

Health Science concentration: For students pursuing a career in the medical professions. Beyond the physics core curriculum, students take additional courses in biology and chemistry. The curriculum has been chosen to satisfy medical school and other health-oriented program requirements.

Quantitative Finance concentration: Prepares students for careers in finance and investing. Contains additional coursework in finance and computer science, with a capstone course in econophysics.

Applied/Computational concentration: For students interested in pursuing a career in engineering or computer/information systems. Students will take an applied physics course and the full suite of introductory computer science courses.

Core Requirements: 54 credits

Required lower division courses: 16 credits

- PX 2200 - Physics 1, Calculus-based Credits: 3
- PX 2201 - Physics 1, Calculus-based lab Credits: 1
- PX 2300 - Physics 2, Calculus-based Credits: 3
- PX 2301 - Physics 2, Calculus-based Lab Credits: 1
- CP 2120 - Principles of Chemistry I Credits: 3

- CP 2121 - Principles of Chemistry Lab I Credits: 1
- CP 2130 - Principles of Chemistry II Credits: 3
- CP 2131 - Principles of Chemistry Lab II Credits: 1

Required upper division courses: 26 credits

- PX 3000 - Waves and Modern Physics Credits: 3
- PX 3100 - Mathematical Methods in Physics Credits: 3
- PX 3200 - Computational Physics Credits: 3
- PX 3600 - Mechanics Credits: 3
- PX 3700 - Electromagnetism Credits: 3
- PX 3900 - Experimental Physics and Electronics Lab Credits: 3
- PX 4100 - Thermodynamics Credits: 3
- PX 4200 - Quantum Mechanics Credits: 3
- PX 5500 - Senior Seminar 1 Credits: 1
- PX 5600 - Senior Seminar 2 Credits: 1

Required math courses: 12 credits

- MA 2310 - Calculus & Analytic Geometry I Credits: 4
- MA 2320 - Calculus & Analytic Geometry II Credits: 4
- MA 3330 - Calculus and Analytic Geometry III Credits: 4

Concentration requirements: 14 credits minimum

General Physics (14 credits)

Two additional advanced physics courses (PX 3000 or above), 6 credits

Two aditional advanced math courses, 8 credits

- MA 3160 - Linear Algebra Credits: 4
- MA 4360 - Differential Equations Credits: 4

  (or other MA 4000 level or above)

Health Sciences (21 credits)

- BS 2400 - Basic Biosciences I Credits: 3
- BS 2401 - Basic Biosciences I Lab Credits: 1

- BS 2410 - Basic Biosciences II Credits: 3
- BS 2411 - Basic Biosciences II Lab Credits: 1
- CP 3300 - Organic Chemistry I Credits: 3
- CP 3302 - Organic Chemistry Lab I Credits: 2
- CP 3310 - Organic Chemistry II Credits: 3
- CP 3312 - Organic Chemistry Lab II Credits: 2
- CP 4490 - Biochemistry for Life Sciences Credits: 3

  OR

- CP 4510 - Biochemistry I Credits: 3

Quantitative Finance (31 credis)

- CS 2510 - Computer Programming I Credits: 4
- MA 2000 - Applied Statistics Credits: 4
- MA 3160 - Linear Algebra Credits: 4
- MA 4360 - Differential Equations Credits: 4
- BU 4762 - Financial Management l Credits: 4
- BU 5740 - Investment Management and Security Analysis Credits: 4
- BU 5789 - Futures and Options Markets Credits: 4
- PX 5100 - Econophysics Credits: 3

Applied/Computational (19 credits)

- CS 2510 - Computer Programming I Credits: 4
- CS 2511 - Computer Programming II Credits: 4
- CS 3810 - Data Structures and Algorithms Credits: 4
- MA 3030 - Discrete Mathematics Credits: 4
- PX 4600 - Applied Mechanics Credits: 3

Liberal Education

Refer to the Liberal Education curriculum for specific requirements.

General Electives

In consultation with Academic Advisor